

N 418

Seat No.

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2024 VII 26 1100 – N 418 – SCIENCE AND TECHNOLOGY (72) – PART II (E)
(REVISED COURSE)

Time : 2 Hours

(Pages 11)

Max. Marks : 40

Note :— (i) *All questions are compulsory.*

(ii) *Use of calculator is not allowed.*

(iii) *The numbers to the right of the questions indicate full marks.*

(iv) *In case of MCQs [Q. No. 1 (A)] only the first attempt will be evaluated and will be given credit.*

(v) *For every MCQ, the correct alternative (A), (B), (C) or (D) with subquestion number is to be written as an answer.*

For e.g. (i) (A) (ii) (B) (iii) (C)

(vi) *Scientifically correct, labelled diagrams should be drawn wherever necessary.*

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1. (A) Choose *correct* option and write its number for the following questions : 5

(i) Evolution means 1

(A) Internal change

(B) Gradual change

(C) External change

(D) Quick change

(ii) One of the rare species is 1

(A) Lesser florican

(B) Red panda

(C) Tiger

(D) Shekhru

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- (iii) The machine which converts the kinetic energy of wind to electrical energy is called 1
- (A) Solar cell
- (B) Thermal
- (C) Generator
- (D) Wind turbine
- (iv) Chemically, vinegar is 1
- (A) Acetic acid
- (B) Amino acid
- (C) Hydrochloric acid
- (D) Gluconic acid
- (v) We get kcal energy per gram of proteins. 1
- (A) 4
- (B) 3
- (C) 2
- (D) 1

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(B) Solve the following questions : **5**

(i) Find odd man out : **1**

Starfish, Snail, Frog, Amoeba

(ii) Match the correct pairs : **1**

Column 'A'

Column 'B'

(1) Biotic components

(a) Soil

(b) Plant

(c) Sunlight

(iii) Write the correlation : **1**

White revolution : Dairy :: Blue revolution :

(iv) Observe the diagram and identify the phylum : **1**



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- (v) State, whether given statement is right or wrong : 1

Peer group influence is stronger in case of adolescents.

2. (A) Give scientific reason (any *two*) : 4

- (i) Fibers are one of the important nutrients.
- (ii) Though tortoise lives on land as well as in water, it cannot be included in class-Amphibia.
- (iii) Hydroelectric energy, solar energy and wind energy are called renewable energies.

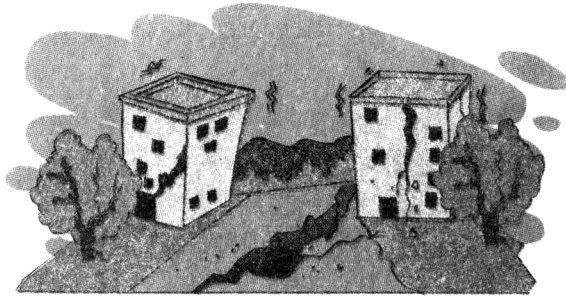
- (B) Answer the following questions (any *three*) : 6

- (i) Write down *four* benefits of Biotechnology.
- (ii) Distinguish between Sexual reproduction and Asexual reproduction.

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(iii) Observe the following diagrams and complete table :



‘A’



‘B’

Disaster



Precaution



Snake Bite



Primary

Stage



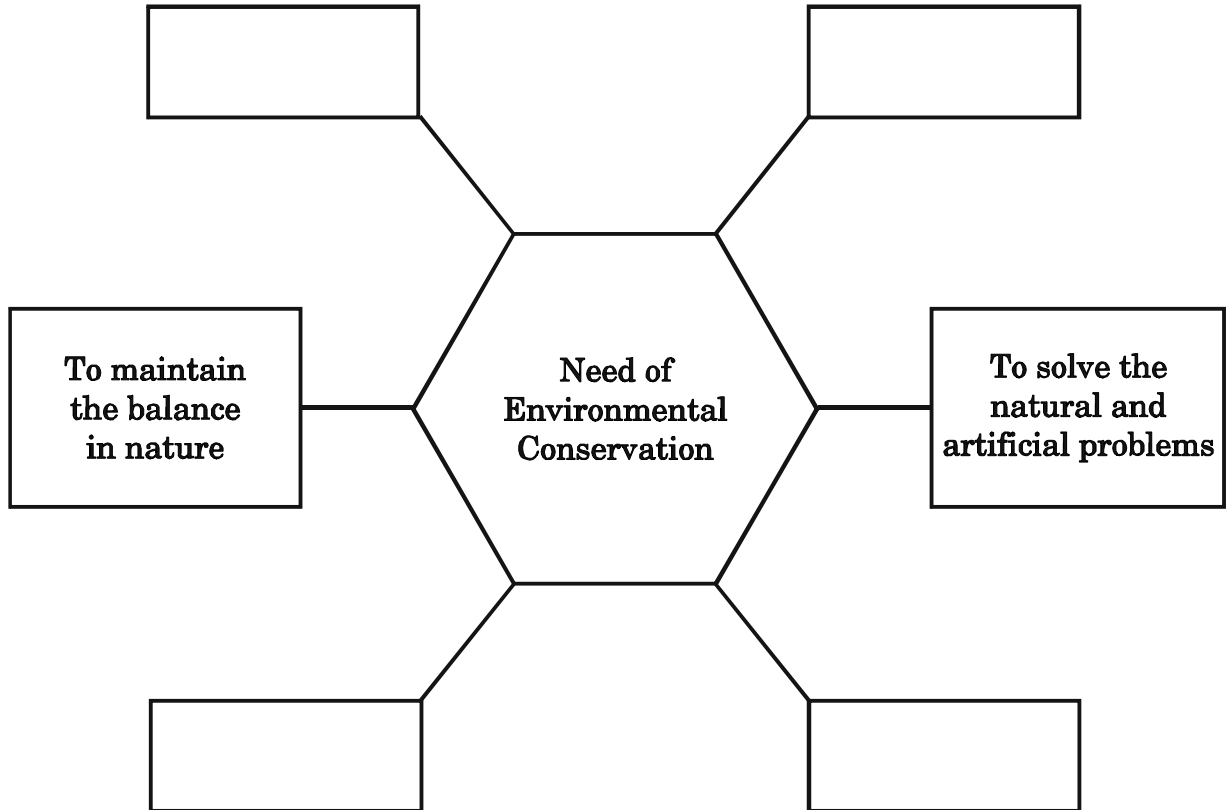
(1)

(2)

(iv) Write down any *two* features of Industrial Microbiology.

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(v) Complete the following flow chart :



3. Solve the following questions (any *five*) :

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(i) Write any *three* characteristic features of class “Aves”.

(ii) (a) What are vestigial organs ?

(b) Write *two* names of some vestigial organs in human body.

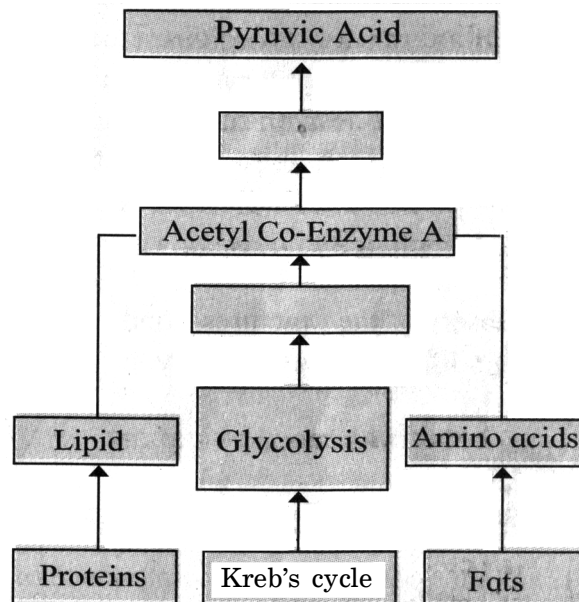
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(c) Write the names of those animals in whom same organs are used ? Give *two* examples.

(iii) How energy is formed from oxidation of carbohydrates, fats and proteins ?

Correct the diagram given below :

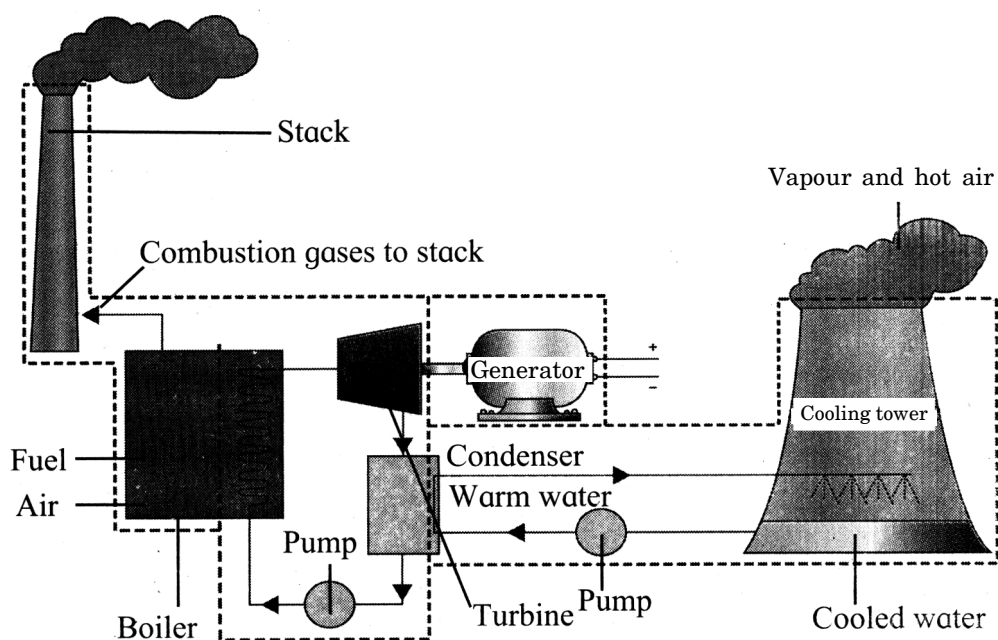


(iv) What is the meaning of the following symbols ? Write your role accordingly.



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(v) Observe the diagram and answer the following questions :



(a) Which type of energy is created in the above diagram ?

(b) Which type of fuel is used in power plant ?

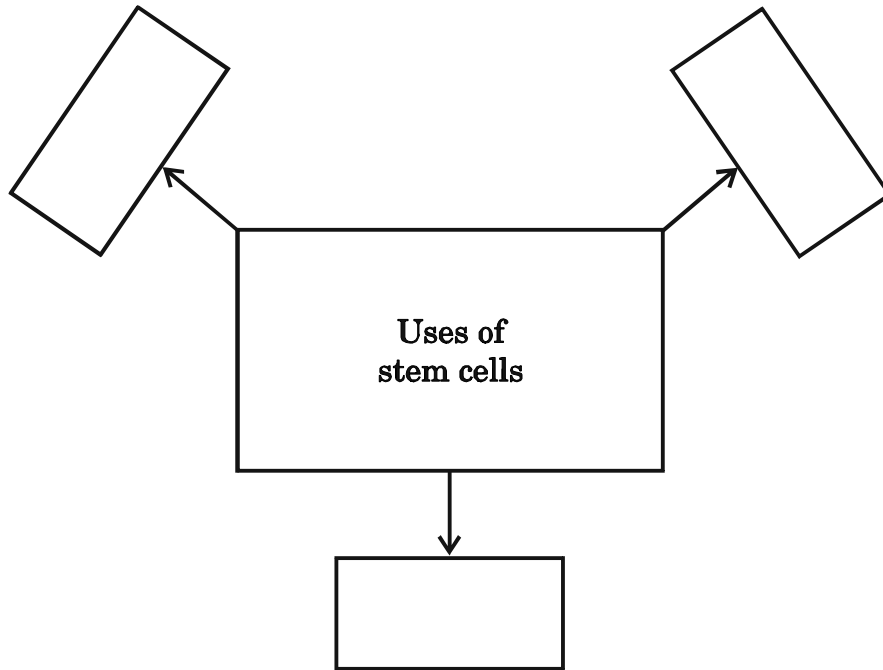
(c) State *one* problem created in such type of power plant.

(vi) Explain the importance of Biopesticides in organic farming.

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- (vii) Complete the following conceptual picture and write its uses in treatment of various diseases :



- (viii) Which are *three* objectives of disaster management ?

4. Solve the following questions (any *one*) : 5
- (i) (a) Draw well labelled diagram of female reproductive system. 2
- (b) Which type of hormones are secreted in female reproductive system ? 1
- (c) When does menopause period stop ? 1
- (d) How many ovaries are there in female foetus at the time of birth ? 1

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(ii) Give *two* examples of each :

- (a) Hobbies to reduce stress. 1
- (b) Diseases endangering the social health. 1
- (c) Physical problems arising due to excessive use of mobile phones. 1
- (d) Activities under the jurisdiction of cyber crime laws. 1
- (e) Write problems due to alcoholism. 1